

Module1: WIRELESS CHANNELS

INTRODUCTION TO WIRELESS COMMUNICATION

Wireless:

- ✓ Wireless is used to describe types of devices and technologies that use space as a signal propagating medium, and are not connected by a wire or cable.
- ✓ Wireless communication may be defined as the transmission of user information without the use of wires.
- ✓ The user information could be in the form human voice, digital data, e-mail messages, video and other services.
- ✓ The wireless communications covering radio, television,, radar, satellite, wireless and mobile, cellular and other wireless networks.

Advantages of wireless communications:

1. **Mobility:** The users have freedom to move about while remaining connected, as compared with the network with its coverage area.
2. **Increased reliability:** Use of wireless technology eliminate cable failures, so overall reliability
3. **Ease of installation:** Wireless communications and networks make it easier for any office to be modified with new cubicles or furniture, without worrying about providing network connectivity through cables.
4. **Rapid disaster recovery:** Accidents may happen due to fire, etc., the organization hot prepared to recover such natural disasters. So, disaster recovery plan is must for business.
5. **Low cost:** Eliminating the need to install cabling and using wireless communications results in significant cost savings.

Disadvantages of wireless communications:

1. **Radio signal interference:** Signals from other wireless devices can disturb its radio transmission.
2. **Security:** Data transmitted between the wireless device and the access point can also be encrypted in such a way that only the intended recipient can decode the message.
3. **Health hazards:** High powered levels of RF energy can produce biological damage.
4. Radio transmitters in wireless communications devices emit radio frequency (RF) energy cause adverse health effects.

Wireless network generations:

The cellular systems have been classified into three generations

- i) First generation analog cellular systems
- ii) Second generation digital cellular systems
- iii) Third generation digital

cellular systems 1G analog cellular systems:

- ✓ Amps (Advance Mobile Phone System) and ETACS (Enhanced Total Access Communication System)

2G digital cellular systems:

- ✓ IS-136, GSM (Global System for Mobile or Group Special for Mobile) and PDC (Personal Digital Cellular)

3G digital cellular systems:

- ✓ It aims to combine telephony, internet and multimedia into a single device.

OUTDOOR PROPAGATION MODELS

Based on the coverage area, the Outdoor propagation environment may be divided into three categories

- Propagation in Macro cells
- Propagation in Micro cells

Outdoor radio transmission takes place over an irregular terrain. The terrain profile must be taken into consideration for estimating the path loss.

e.g. trees buildings and hills must be taken into consideration

Longley-Ryce Model:

The Longley-Ryce model is applicable to point-to-point communication systems in the frequency range from 40 MHz to 100 GHz, over different kinds of terrain. The median transmission loss is predicted using the path geometry of the terrain profile and the refractivity of the troposphere. Geometric optics techniques (primarily the 2-ray ground reflection model) are used to predict signal strengths within the radio horizon. Diffraction losses over isolated obstacles are estimated using the Fresnel-Kirchoff knife-edge models. Forward scatter theory is used to make troposcatter predictions over long distances.

The Longley-Ryce method operates in two modes. When a detailed terrain path profile is available, the path-specific parameters can be easily determined and the prediction is called a point-to-point mode prediction. On the other hand, if the terrain path profile is not available, the Longley-Ryce method provides techniques to estimate the path-specific parameters, and such a prediction is called an area mode prediction.

Okumura Model:

Okumura's model is one of the most widely used models for signal prediction in urban areas. This model is applicable for frequencies in the range 150 MHz to 1920 MHz (although it is typically extrapolated up to 3000 MHz) and distances of 1 km to 100 km. It can be used for base station antenna heights ranging from 30 m to 1000 m. Okumura developed a set of curves giving the median attenuation relative to free space (A_{mu}), in an urban area over a quasi-smooth terrain with a base station effective antenna height (h_{te}) of 200 m and a mobile antenna height (h_{re}) of 3 m. These curves were developed from extensive measurements using vertical omni-directional antennas at both the base and mobile, and are plotted as a function of frequency in the range 100 MHz to 1920 MHz and as a function of distance from the base station in the range 1 km to 100 km. To determine path loss using Okumura's model, the free space path loss between the points of interest is first determined, and then the value of $A_{mu}(f, d)$ (as read from the curves) is added to it along with correction factors to account for the type of terrain. The model can be expressed as

$$L_{50} \text{ (dB)} = L_F + A_{mu}(f, d) - G_{(te)} - G_{(re)} - G_{AREA}$$

where L_{50} is the 50th percentile (i.e., median) value of propagation path loss, L_F is the free space propagation loss, A_{mu} is the median attenuation relative to free space, $G(h_{te})$ is the base station antenna height gain factor, $G(h_{re})$ is the mobile antenna height gain factor, and $G_{A R E A}$ is the gain due to the type of environment. Note that the antenna height gains are strictly a function of height and have nothing to do with antenna patterns.

$$G(h_{te}) = 20\log\left(\frac{h_{te}}{200}\right) \quad 1000 \text{ m} > h_{te} > 30 \text{ m}$$

$$G(h_{re}) = 10\log\left(\frac{h_{re}}{3}\right) \quad h_{re} \leq 3 \text{ m}$$

$$G(h_{re}) = 20\log\left(\frac{h_{re}}{3}\right) \quad 10 \text{ m} > h_{re} > 3 \text{ m}$$

Hata Model:

The Hata model [Hat90] is an empirical formulation of the graphical path loss data provided by Okumura, and is valid from 150 MHz to 1500 MHz. Hata presented the urban area propagation loss as a standard formula and supplied correction equations for application to other situations. The standard formula for median path loss in urban areas is given by

$$L_{50}(\text{urban}) (\text{dB}) = 69.55 + 26.16\log f_c - 13.82\log h_{te} - a(h_{re}) + (44.9 - 6.55\log h_{te})\log d$$

where f_c is the frequency (in MHz) from 150 MHz to 1500 MHz, h_{te} is the effective transmitter (base station) antenna height (in meters) ranging from 30 m to 200 m, h_{re} is the effective receiver (mobile) antenna height (in meters) ranging from 1 m to 10 m, d is the T-R separation distance (in km), and $a(h_{re})$ is the correction factor for effective mobile antenna height which is a function of the size of the coverage area. For a small to medium sized city, the mobile antenna correction factor is given by

$$a(h_{re}) = (1.1\log f_c - 0.7)h_{re} - (1.56\log f_c - 0.8) \text{ dB}$$

for a large city, it is given by

$$a(h_{re}) = 8.29(\log 1.54h_{re})^2 - 1.1 \text{ dB for } f_c \leq 300 \text{ MHz} \quad a(h_{re}) = 3.2(\log 11.75h_{re})^2 - 4.97 \text{ dB for } f_c \geq 300 \text{ MHz}$$

To obtain the path loss in a suburban area the standard Hata formula in equations are modified as

$$L_{50}(\text{dB}) = L_{50}(\text{urban}) - 2[\log(f_c/28)]^2 - 5.4$$

and for path loss in open rural areas, the formula is modified as

$$L_{50}(\text{dB}) = L_{50}(\text{urban}) - 4.78(\log f_c)^2 + 18.33\log f_c - 40.94$$

PCS Extension to Hata Model

The European Co-operative for Scientific and Technical research (EURO-COST) formed the COST-231 working committee to develop an extended version of the Hata model. COST-231 proposed the following formula to extend Hata's model to 2 GHz. The proposed model for path loss is [EUR91]

$$L_{50}(\text{urban}) = 46.3 + 33.9 \log f_c - 13.82 \log h_{te} - a(h_{re}) + (44.9 - 6.55 \log h_{te}) \log d + C_M \quad (3.87)$$

where $a(h_{re})$ is defined in equations (3.83), (3.84.a), and (3.84.b) and

$$C_M = \begin{cases} 0 \text{ dB} & \text{for medium sized city and suburban areas} \\ 3 \text{ dB} & \text{for metropolitan centers} \end{cases} \quad (3.88)$$

The COST-231 extension of the Hata model is restricted to the following range of parameters:

$$\begin{aligned} f &: 1500 \text{ MHz to } 2000 \text{ MHz} \\ h_{te} &: 30 \text{ m to } 200 \text{ m} \\ h_{re} &: 1 \text{ m to } 10 \text{ m} \\ d &: 1 \text{ km to } 20 \text{ km} \end{aligned}$$

Walfisch and Bertoni Model

A model developed by Walfisch and Bertoni [Wal88] considers the impact of rooftops and building height by using diffraction to predict average signal strength at street level. The model considers the path loss, S , to be a product of three factors.

$$S = P_0 Q^2 P_1 \quad (3.89)$$

where P_0 represents free space path loss between isotropic antennas given by

$$P_0 = \left(\frac{\lambda}{4\pi R} \right)^2 \quad (3.90)$$

The factor Q^2 gives the reduction in the rooftop signal due to the row of buildings which immediately shadow the receiver at street level. The P_1 term is

based upon diffraction and determines the signal loss from the rooftop to the street.

In dB, the path loss is given by

$$S \text{ (dB)} = L_0 + L_{rts} + L_{ms} \quad (3.91)$$

where L_0 represents free space loss, L_{rts} represents the “rooftop-to-street diffraction and scatter loss”, and L_{ms} denotes multiscreen diffraction loss due to the rows of buildings [Xia92]. Figure 3.25 illustrates the geometry used in the Walfisch Bertoni model [Wal88], [Mac93]. This model is being considered for use by ITU-R in the IMT-2000 standards activities.

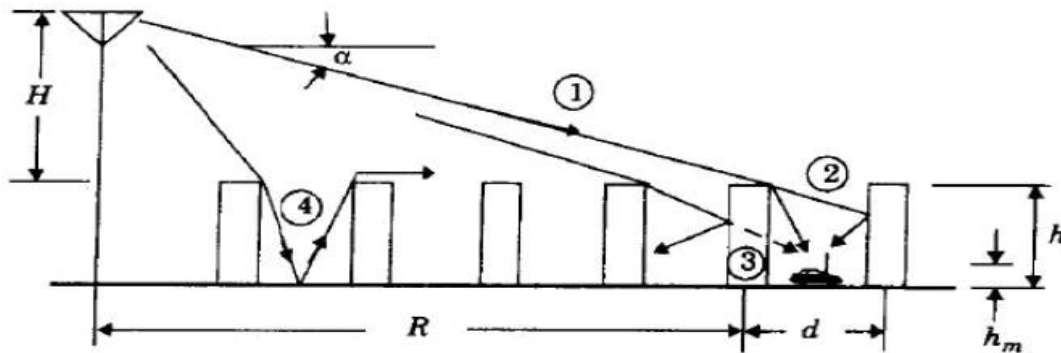


Figure 3.25

Propagation geometry for model proposed by Walfisch and Bertoni [From [Wal88] © IEEE].

Wideband PCS Microcell Model

Work by Feuerstein, et.al. in 1991 used a 20 MHz pulsed transmitter at 1900 MHz to measure path loss, outage, and delay spread in typical microcellular systems in San Francisco and Oakland. Using base station antenna heights of 3.7 m, 8.5 m, and 13.3 m, and a mobile receiver with an antenna height of 1.7 m above ground, statistics for path loss, multipath, and coverage area were developed from extensive measurements in line-of-sight (LOS) and obstructed (OBS) environments [Feu94]. This work revealed that a 2-ray ground reflection model (shown in Figure 3.7) is a good estimate for path loss in LOS microcells, and a simple log-distance path loss model holds well for OBS microcell environments.

For a flat earth ground reflection model, the distance d_f at which the first Fresnel zone just becomes obstructed by the ground (first Fresnel zone clearance) is given by

$$d_f = \frac{1}{\lambda} \sqrt{(\Sigma^2 - \Delta^2)^2 - 2(\Sigma^2 + \Delta^2) \left(\frac{\lambda}{2}\right)^2 + \left(\frac{\lambda}{2}\right)^4} \quad (3.92.a)$$

$$= \frac{1}{\lambda} \sqrt{16h_t^2 h_r^2 - \lambda^2 (h_t^2 + h_r^2) + \frac{\lambda^4}{16}}$$

For LOS cases, a double regression path loss model that uses a regression breakpoint at the first Fresnel zone clearance was shown to fit well to measurements. The model assumes omnidirectional vertical antennas and predicts average path loss as

$$PL(d) = \begin{cases} 10n_1 \log(d) + p_1 & \text{for } 1 < d < d_f \\ 10n_2 \log(d/d_f) + 10n_1 \log d_f + p_1 & \text{for } d > d_f \end{cases} \quad (3.92.b)$$

where p_1 is equal to $PL(d_0)$ (the path loss in decibels at the reference distance of $d_0 = 1$ m), d is in meters and n_1, n_2 are path loss exponents which are a function of transmitter height, as given in Figure 3.26. It can easily be shown that at 1900 MHz, $p_1 = 38.0$ dB.

For the OBS case, the path loss was found to fit the standard log-distance path loss law of equation (3.69.a)

$$PL(d) [dB] = 10n \log(d) + p_1 \quad (3.92.c)$$

For the OBS case, the path loss was found to fit the standard log-distance path loss law of equation (3.69.a)

INDOOR PROPAGATION MODELS

With the advent of Personal Communication Systems (PCS), there is a great deal of interest in characterizing radio propagation inside buildings. The indoor radio channel differs from the traditional mobile radio channel in two aspects - the distances covered are much smaller, and the variability of the environment is much greater for a much smaller range of T-R separation distances. It has been observed that propagation within buildings is strongly influenced by specific features such as the layout of the building, the construction materials, and the building type. This section outlines models for path loss within buildings.

Indoor radio propagation is dominated by the same mechanisms as outdoor: reflection, diffraction, and scattering. However, conditions are much more variable. For example, signal levels vary greatly depending on whether interior doors are open or closed inside a building. Where antennas are mounted also impacts large-scale propagation. Antennas mounted at desk level in a partitioned office receive vastly different signals than those mounted on the ceiling. Also, the smaller propagation distances make it more difficult to insure far-field radiation for all receiver locations and types of antennas.

Partition Losses (same floor):

Buildings have a wide variety of partitions and obstacles which form the internal and external structure. Houses typically use a wood frame partition with plaster board to form internal walls and

have wood or non-reinforced concrete between floors. Office buildings, on the other hand, often have large open areas (open plan) which are constructed by using moveable office partitions so that the space may be reconfigured easily, and use metal reinforced concrete between floors. Partitions that are formed as part of the building structure are called hard partitions, and partitions that may be moved and which do not span to the ceiling are called soft partitions. Partitions vary widely in their physical and electrical characteristics, making it difficult to apply general models to specific indoor installations.

Partition Losses between Floors:

The losses between floors of a building are determined by the external dimensions and materials of the building, as well as the type of construction used to create the floors and the external surroundings. Even the number of windows in a building and the presence of tinting (which attenuates radio energy) can impact the loss between floors. It can be seen that for all three buildings, the attenuation between one floors of the building is greater than the incremental attenuation caused by each additional floor. After about five or six floor separations, very little additional path loss is experienced.

Log-distance Path Loss Model:

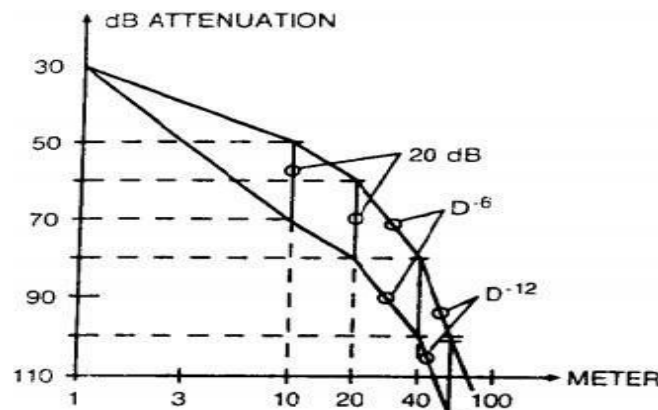
According to this model the received power at distance d is given by,

$$PL(dB) = PL(d_0) + 10n \log\left(\frac{d}{d_0}\right) + X_\sigma$$

The value of n varies with propagation environments. The value of n is 2 for free space. The value of n varies from 4 to 6 for obstruction of building, and 3 to 5 for urban scenarios. The important factor is to select the correct reference distance d_0 . For large cell area it is 1 Km, while for micro-cell system it varies from 10m-1m.

Ericsson multiple breakpoint model:

It was obtained by measurements in a multiple floor office building. It has 4 breakpoints and considers both an upper and lower bound on path loss. It assumes that there is 30dB attenuation at $d_0 = 1m$ which is accurate for $f = 900MHz$ & unity gain antennas.



Small Scale Fading and Multipath

- ✓ The term small-scale fading or simply *fading*, means rapid fluctuations of the amplitudes, phases, or multipath delays of a radio signal over a short period of time or short travel distance, so that the large scale pathloss effects may be ignored.
- ✓ This might be so severe that large scale radio propagation loss effects might be ignored.
- ✓ Fading is caused by interference between two or more versions of the transmitted signal which arrive at the receiver at slightly different times. These waves are called ***multipath waves***.

Small-Scale Multipath Propagation:

- In principle, the following are the main multipath effects:
 1. Rapid changes in signal strength over a small travel distance or time interval.
 2. Random frequency modulation due to varying Doppler shifts on different multipath signals.
 3. Time dispersion or echoes caused by multipath propagation delays.

Factors Influencing Fading:

The following physical factors influence small-scale fading in the radio propagation channel:

(1) Multipath propagation

- The reflecting objects and scatterers in the channel creates a constantly changing environment.
- This changing environment dissipates the signal energy in amplitude, phase and time.
- This effect results in multiple versions of the transmitted signal that arrive at the receiving antenna.
- These received signals displace with respect to one another in time and spatial orientation.
- The random phase and amplitudes of the different multipath components cause fluctuations in signal strength, includes small scale fading, signal distortion or both.
- Multipath causes signal smearing (***spreading***) due to ISI.

(2) Speed of the mobile

- The relative motion between the base station and the mobile results in random frequency modulation due to different Doppler shifts on each of the multipath components.
- Doppler shift will be positive (*i.e., apparent receiving frequency is increased*) or negative, it depends on the mobile moving toward or away from the base station respectively.

(3) Speed of surrounding objects

- If objects in the radio channel are in motion, they induce a time varying Doppler shift on multipath components.
- If the surrounding objects move at a greater rate than the mobile, then this effect dominates fading.

(4) Transmission Bandwidth of the signal

- If the transmitted radio signal bandwidth is greater than the “bandwidth” of the multipath channel (quantified by coherence bandwidth), the received signal will be distorted.
- But, the received signal strength will not fade much over a local area.

1.3. PARAMETERS OF MOBILE MULTIPATH CHANNELS

- ✓ Power delay profiles are averaging instantaneous power delay profile over a local area to determine an average small-scale power delay profile.

- ✓ Parameters that describes the time varying nature of the channel in a small scale region are
 - **Doppler spread**
 - **Coherence time**

1.3.1. TIME DISPERSION PARAMETERS

- ✓ In multipath phenomenon, the time difference between the arrival moment of the first multipath component and the last one is called **delay spread**.
 - ✓ Some parameters are used to quantify the multipath channel.
 - ✓ These parameters are used to compare different multipath channels and to develop design guidelines for wireless systems.
 - ✓ The following multipath parameters are used to quantify the time dispersive properties of wide band multipath channels:
 - a. Mean excess delay ($\bar{\tau}$)
 - b. RMS delay spread (σ_{τ})
 - c. Maximum Excess Delay (X dB)
- (frequently used properties)

a. Mean excess delay

- ✓ The mean excess delay is the first moment of the power delay profile (PDP).
- ✓ It is expressed as

$$\bar{\tau} = \frac{\sum_k a_k^2 \tau_k}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k) \tau_k}{\sum_k P(\tau_k)}$$

b. RMS delay spread

- ✓ This is the most important single measure for the delay times extent of a multipath delay channel.
- ✓ This parameter calculates the standard deviation value of the delay of reflections.
- ✓ The standard deviation value will be weighted proportional to the energy in the reflected waves.
- ✓ The *rms* delay spread is the square root of the second central moment of the power delay profile.

$$\sigma_{\tau} = \sqrt{\overline{\tau^2} - (\overline{\tau})^2}$$

$$\overline{\tau^2} = \frac{\sum a_k^2 \tau_k^2}{\sum a_k^2} = \frac{\sum P(\tau_k) \tau_k^2}{\sum P \tau_k}$$

where,

σ_{τ} → *rms* delay spread

a_k → Amplitude

$P(\tau_k)$ → Relative power levels of the individual multipath components

τ_k → Excess delay

d. Maximum excess delay

- ✓ The maximum excess delay (X dB) of the power delay profile is defined to be the time delay during which multipath energy falls to X dB below the maximum.
- ✓ The maximum excess delay is defined $\tau_x - \tau_0$ as
Where, τ_0 → First arriving signal
 τ_x → Maximum delay at which a multipath component is within X dB of the strongest multipath signal
- ✓ Maximum excess delay is sometimes called the **excess delay spread**.
- ✓ The maximum excess delay defines the temporal extent of multipath that is above a particular threshold.
- ✓ In all cases, must be specified with a threshold that relates the multipath noise floor to maximum received multipath component.

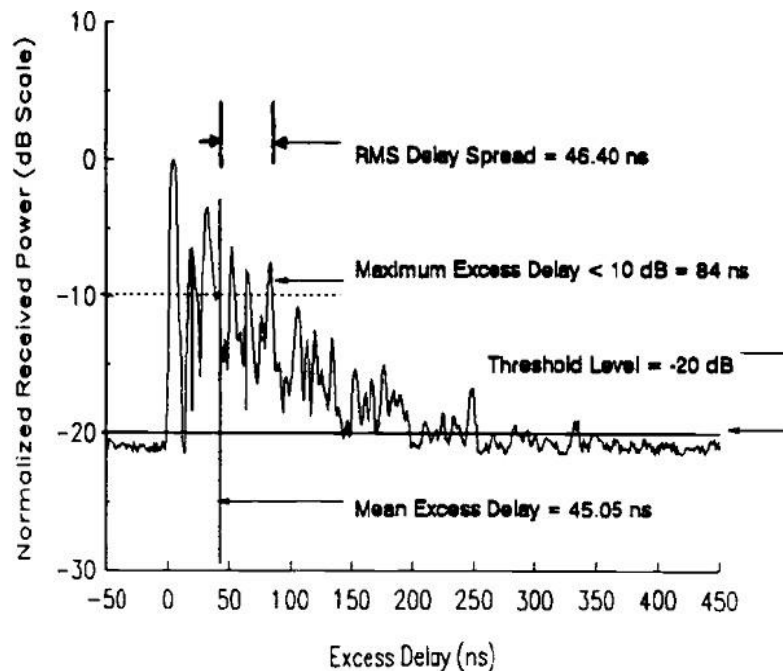


Figure 1.13: Measured multipath power delay profiles

1.3.2. COHERENCE BANDWIDTH

- ✓ The maximum frequency separation for which the signals are still strongly correlated is called coherence bandwidth (B_c).
- ✓ Coherence bandwidth, B_c , is derived from the rms delay spread.
- ✓ Coherence bandwidth is a statistical measure of the range of frequencies over which the channel is considered to be 'flat'.
- ✓ If frequency correlation function is above 0.9, then the coherence bandwidth is $B_c \approx \frac{1}{50\sigma_\tau}$
- ✓ If frequency correlation function is above 0.5, then the coherence bandwidth is $B_c \approx \frac{1}{5\sigma_\tau}$
- ✓ If the signal bandwidth larger than B_c is transmitted through the channel, it will subject to Frequency selective distortion.
The channel will be referred as a frequency selective channel.
- ✓ If the signal bandwidth is larger than B_c , it will experience amplitude attenuation only with no distortion.
- ✓ This channel will be referred as a frequency non-selective fading channel.

1.3.3. DOPPLER SPREAD AND COHERENCE TIME

- ✓ Doppler spread and Coherence time are parameters which describe the time dispersive nature of the channel in a local area.
- ✓ Also, describes the time varying nature of the channel in a small-scale region.
- ✓ They do not give information about the time varying nature of the channel in large scale region.

Doppler spread (B_D)

- ✓ Doppler spread B_D is a measure of the spectral broadening caused by the time rate of change of the mobile radio channel
- ✓ Doppler spread B_D is defined as the range of frequencies over which the received Doppler spectrum is essentially non-zero.
- ✓ When pure dc signal is transmitted, then Doppler spectrum is $f_c, f_c + f_d, f_c - f_d$
where $f_d \rightarrow$ Doppler frequency.
- ✓ It describes the time varying nature frequency dispersiveness of the channel
- ✓ If the baseband signal bandwidth is much greater than B_D , the effects of Doppler spread are negligible at the receiver. This is a slow fading channel.

Coherence time (T_C)

- ✓ Coherence time is the time over which two signals are having strong potential for amplitude correlation.

- ✓ Coherence time is the range of time over which similar fading occurs.
- ✓ The Doppler spread and coherence time are inversely proportional to one another.

$$\text{Coherence Time} = \frac{1}{\text{Doppler Spread}}$$

- ✓ Thus, if the transmitter, receiver, or the intermediate objects move very fast, the Doppler spread is large and the coherence time is small, i.e., the channel changes fast.
- ✓ That is,

$$T_c \approx \frac{1}{f_m}$$

Where,

$$f_m \rightarrow \text{Maximum Doppler Shift, } f_m = \frac{v}{\lambda}$$

- ✓ The time over which the time correlation > 0.5 , then the coherence time is

$$T_c \approx \frac{9}{16\pi f_m}$$

- ✓ Equation is valid for Rayleigh fading channels.
- Coherence time is defined as the geometric mean

$$T_c = \sqrt{\frac{9}{16\pi f_m^2}} = \frac{0.423}{f_m}$$

- ✓ If the reciprocal bandwidth of the baseband signal is greater than the coherence time of the channel, then the channel will change during the transmission of the baseband message, thus causing distortion at the receiver.

1.4. TYPES OF SMALL SCALE FADING

- ✓ The type of fading in the signal propagating through a mobile radio channel depends on the nature of the transmitted signal with respect to the characteristics of the channel.
- ✓ Depending on the relation between the signal parameters and channel parameters, different transmitted signals will undergo different types of fading.
- ✓ Eg: Signal parameters – Bandwidth, symbol period, etc Channel parameters – RMS delay and Doppler spread.
- ✓ The time dispersion and frequency dispersion mechanisms in a mobile radio channel lead to four possible effects.
- ✓ The four different types of fading are

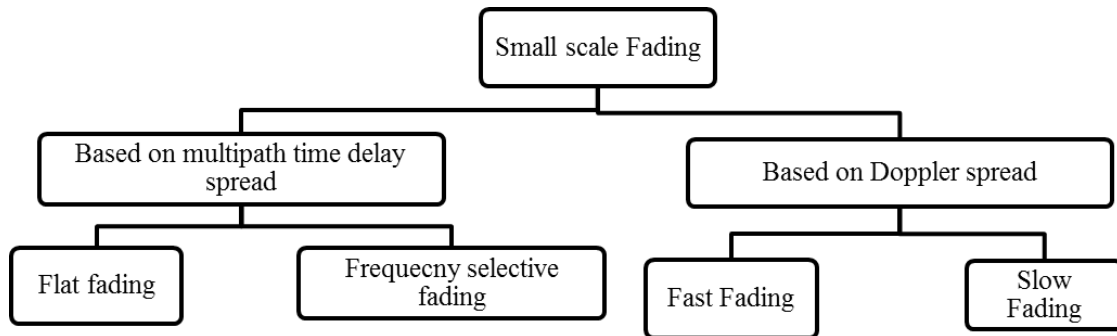
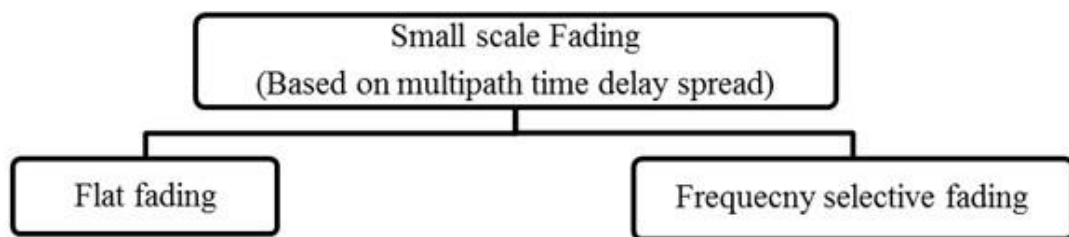


Figure 1.14: Types of small scale fading

1.6.1. FADING EFFECTS DUE TO MULTIPATH TIME DELAY SPREAD

- ✓ Time dispersion due to multipath causes the transmitted signal to undergo either **flat** or **frequency selective fading**.



- Bandwidth of signal < Bandwidth of channel
- Delay spread < Symbol period

- 1. Bandwidth of signal > Bandwidth of channel
- 2. Delay spread > Symbol period

Figure 1.15: Types of small scale fading (Based on Multipath time delay spread)

a. **Flat Fading:**

- ✓ This form of multipath fading affects all the frequencies (almost equally) in the given channel.
- ✓ Flat multipath fading changes the amplitude and rising & falling time of the signal.
- ✓ If mobile radio channel has a constant gain and linear phase response over a bandwidth that is greater than the bandwidth of the transmitted signal, then the received signal will undergo flat fading.

✓ **Channel impulse response**

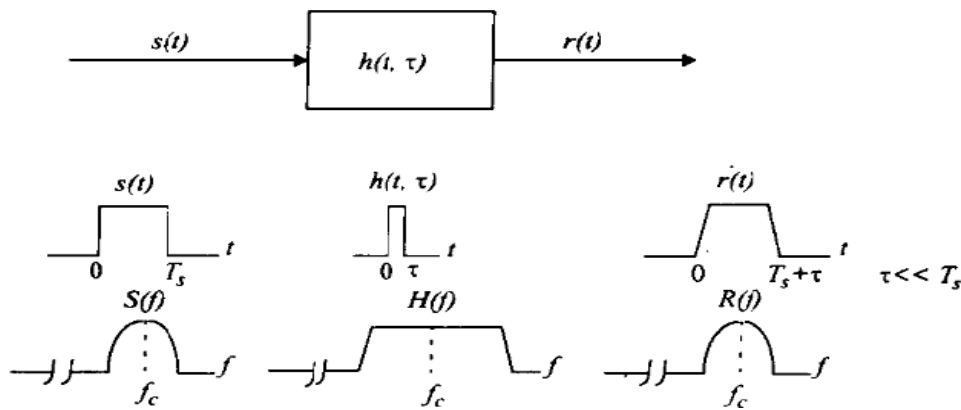
$$\text{Multipath delay spread} \propto \frac{1}{\text{Bandwidth of transmitted message waveform}}$$

✓ **Time characteristics**

- Received signal changes with time due to fluctuations in the gain of the channel caused by multipath.

✓ **Spectral characteristics**

- Multipath structure of the channel is such that the spectral characteristics of the transmitted signal are preserved at the receiver.

➤ **Channel characteristics****Figure 1.16:** Flat fading channel characteristics

- ✓ The amplitude and channel gain varies with time, but the spectrum of the transmission is preserved.
- ✓ Flat fading channels are also known as amplitude varying channels (or) narrowband channels (since, the bandwidth of the applied signal is narrow than the channel bandwidth).
- ✓ Flat fading channels cause deep fades and requires 20 or 30 dB more transmitter power to achieve low bit error rates.
- ✓ Time varying statistics are like Rayleigh flat fading.
- ✓ Rayleigh distribution channels are used to measure variations of amplitude.
- ✓ A signal undergoes flat fading if

$$B_s \ll B_c \text{ and } T_s \gg \sigma_\tau$$

where, $B_s \rightarrow$ Signal Bandwidth.

$B_c \rightarrow$ Coherence bandwidth

$T_s \rightarrow$ Reciprocal bandwidth (symbol period)

$\sigma_\tau \rightarrow$ rms delay spread

b. Frequency Selective Fading

- ✓ Assume the channel possesses a constant-gain and linear phase response over a bandwidth that is smaller than the bandwidth of transmitted signal, then the channel creates frequency selective fading on the received signal.

✓ **Channel impulse response**

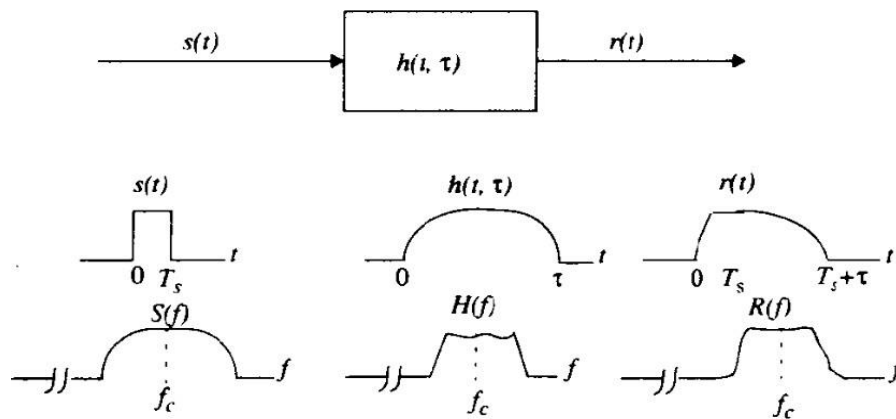
$$\text{Mutipath delay spread} \propto \frac{1}{\text{Bandwidth of transmitted message waveform}}$$

✓ **Time characteristics**

- For this, the received signal includes multiple versions of the transmitted waveform which are attenuated (delayed) and delayed in time. Hence the received signal is distorted.

✓ **Spectral characteristics**

- Frequency selective fading is due to time dispersion of the transmitted symbols within the channel. Thus the channel induces intersymbol interference (ISI).
- The received signal spectrum has greater gains than others.

✓ **Channel characteristics****Figure 1.17:** Frequency selective fading channel characteristics

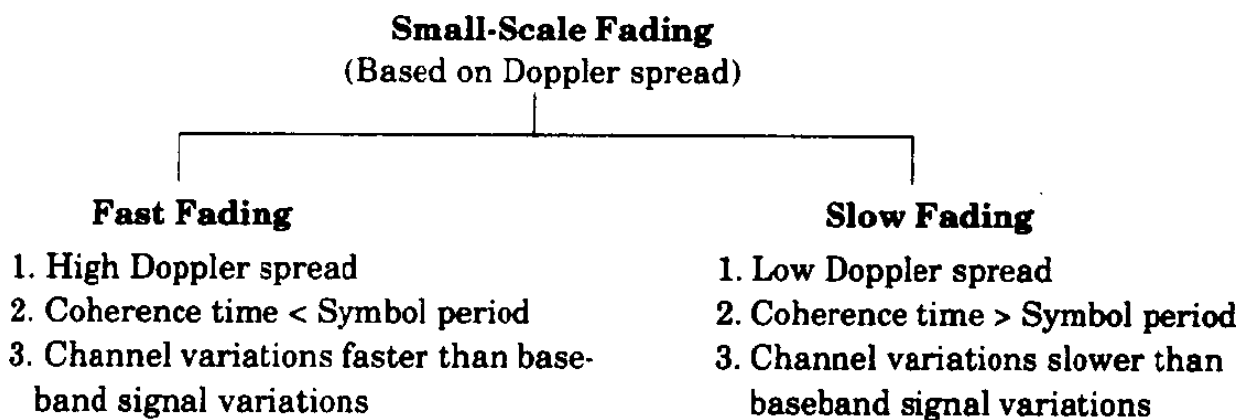
- ✓ The gain is different for different frequency components.
- ✓ Frequency selective fading channels are also known as wideband channels since the bandwidth of the signal $s(t)$ is wider than the bandwidth of the channel impulse response
- ✓ Selective fading lowers less fading effects [Time varying distortions]
- ✓ Fading models are used to measure frequency.
- ✓ For Frequency selective fading, the spectrum $S(f)$ of the transmitted signal bandwidth (B_s) which is greater than the coherence bandwidth (B_c)
- ✓ That is, a signal undergoes frequency selective fading if

$$B_s > B_c \text{ and } T_s > \sigma_\tau,$$

where	B_s	→	Signal Bandwidth.
	B_c	→	Coherence bandwidth
	T_s	→	Reciprocal bandwidth
	σ_τ	→	rms delay spread

1.6.2. FADING EFFECTS DUE TO DOPPLER SPREAD

- ✓ Depending on how rapidly the transmitted baseband signal changes as compared to the rate of change of the channel, a channel may be classified as follows.

**Figure 1.18:** Types of small scale fading (Based on Doppler spread)

a. Fast Fading / Time Selective Fading

- ✓ In a fast fading channel, the channel impulse response changes rapidly within the symbol duration.
- ✓ That is, the coherence time of the channel is smaller than the symbol period of the transmitted signal.
- ✓ This causes Frequency dispersion (also called time selective fading) due to Doppler spreading, which leads to signal distortion.
- ✓ In the frequency domain, signal distortion due to fast fading increases with increasing Doppler spread relative to the bandwidth of the transmitted signal.
- ✓ Therefore, a signal undergoes fast fading if

$$T_s > T_c \text{ and } B_s < B_D$$

- ✓ It should be noted that when a channel is specified as a fast or slow fading channel, it does not specify whether the channel is flat fading or frequency selective in nature.
- ✓ Fast fading only deals with the rate of change of the channel due to motion.
- ✓ In flat fading channel, the impulse response of the flat fading channel is a delta function (no time delay).
- ✓ Hence, a *flat fading, fast fading channel* is a channel in which amplitude of the delta function varying faster than the rate of change of the transmitted baseband signal.
- ✓ In case of a *frequency selective, fast fading channel*, the amplitudes, phases, and time delays of any one of the multipath components vary faster than the rate of change of the transmitted signal.
- ✓ In practice, fast fading only occurs for very low data rates.

b. Slow Fading

- ✓ In a slow fading channel, the channel impulse response changes at a rate much slower than the transmitted baseband signal $s(t)$.
- ✓ In this case, the channel is assumed to be static over one or several reciprocal bandwidth intervals.
- ✓ In the frequency domain, the Doppler spread of the channel is much less than the bandwidth of the baseband signals.
- ✓ Therefore, a signal undergoes slow fading if

$$T_s \ll T_c$$

$$B_s \gg B_D$$

- ✓ The relation between the various multipath parameters and the type of fading experienced by the signal are summarized in Figure.

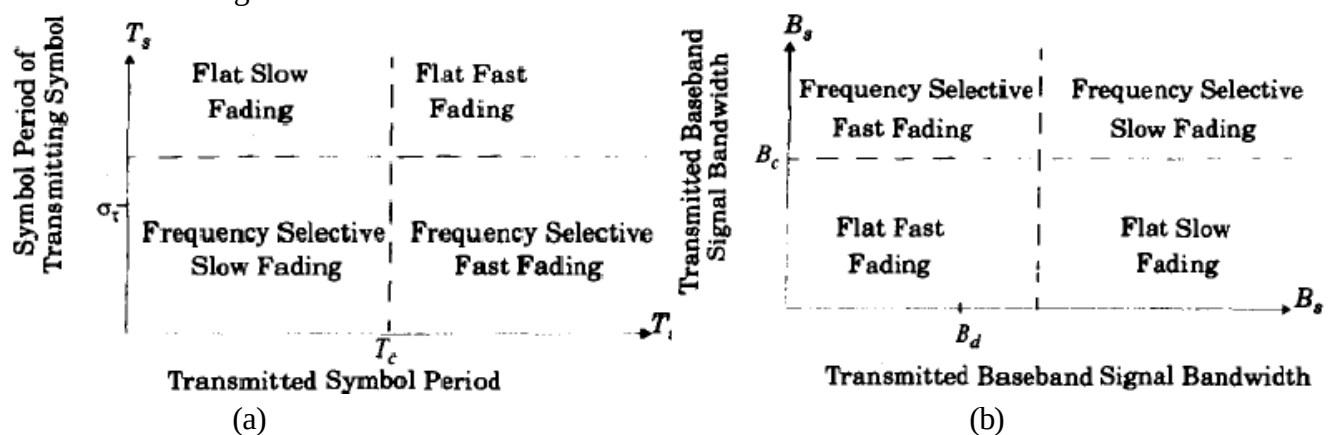


Figure 1.19: Matrix illustrating type of fading experienced by a signal as a function of

(a) symbol period

(b) baseband signal bandwidth

State the difference between small-scale and large-scale propagation.

LARGE-SCALE PROPAGATION	SMALL-SCALE PROPAGATION
redicts mean signal strength for an arbitrary transmitter-receiver (T-R) separation distance are useful in estimating the radio coverage area of a transmitter is called large-scale propagation	Rapid fluctuations of the received signal strength over very short travel distance/short duration are called Small-scale propagation.
As the mobile moves away from transmitter over large distances, the local average received signal will gradually decrease	As the mobile moves away from transmitter over small distances, , the received signal may fluctuate, giving rise to small scale fading
The local average signal is computed by large scale propagation models typically (Computed by averaging signal measurements over measurement track) cellular=> 1 GHz -2 GHz band→ power movement from 1m to 10m.	The received power may vary from [30/40 dB] when the receiver is moved by fraction of wavelength.

Solved Examples

Example 1.1 A mobile is located 10 Km away from a base station and uses a vertical $\lambda/4$ monopole antenna with a gain of 3 d B to receive cellular radio signals. The E-field at 1 km from the transmitter is measured to be 10^{-3} V/m. The carrier frequency used for this system is 1000 MHz.

(a) Find the length and effective aperture of the receiving antenna

(b) Find the received power at the mobile using the two-ray ground reflection model assuming h_t is 50 m and h_r is 1.5 m above ground.

Given data

Transmitter-receiver distance = 10 Km

E-field at 1 Km = 10^{-3} V/m

Frequency of operation = f = 1000 MHz

Solution:

(a) Length of antenna $L = \frac{\lambda}{4}$

$$\begin{aligned} \text{Wavelength } \lambda &= \frac{c}{f} \\ &= \frac{3 \times 10^8}{1000 \times 10^6} = 0.3 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Length } L &= \frac{\lambda}{4} \\ &= \frac{0.3}{4} = 0.075 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Effective aperture of antenna } A_e &= \frac{G \lambda^2}{4 \pi} \\ &= \frac{(0.3)^2}{4 \pi} = 0.015 \text{ m}^2 \end{aligned}$$

$$\begin{aligned}
 \text{(b) Received Power } P_r(d) &= \frac{|E|^2 G_r \lambda^2}{377 (4 \pi)} \text{ watts} \\
 \text{Electric field } |E| &= \frac{2 E_0 d_0}{d} \left(\frac{2 \pi h_t h_r}{\lambda d} \right) \\
 &= \frac{2 \times 10^{-3} \times 10^3}{10 \times 10^3} \left(\frac{2 \pi (50)(1.5)}{0.3 \times 10 \times 10^3} \right)
 \end{aligned}$$

$$\text{Received Power } P_r = 3.14 \times 10^{11} \text{ watts}$$

Example 1.2 Consider a transmitter which radiates a sinusoidal carrier frequency of 2000 MHz. For a vehicle moving 100 mph, compute the received carrier frequency if the mobile is moving

(a) directly toward the transmitter

(b) away from the transmitter

(c) direction which is perpendicular to the direction of arrival of the transmitted signal.

Given Data:

$$\text{Carrier frequency } f_c = 2000 \text{ MHz}$$

$$\text{Vehicle speed } V = 100 \text{ mph}$$

Solution:

(a) Doppler shift is positive for vehicle is moving towards the transmitter

$$\text{Received frequency } f = f_c + f_d$$

$$= 2000 \times 10^6 + \frac{V}{\lambda}$$

$$= 2000 \times 10^6 + \frac{100 \times 60 \times 60}{0.15}$$

$$= 2002 \text{ MHz}$$

$$f = \frac{c}{\lambda}$$

$$\lambda = \frac{c}{f_c}$$

$$= \frac{3 \times 10^8}{2000 \times 10^6} = 0.15 \text{ m}$$

(b) Doppler shift is negative for vehicle is moving away from the transmitter

$$f = f_c - f_d$$

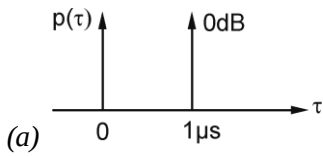
$$= 2000 \times 10^6 - \frac{V}{\lambda}$$

$$= 1997 \text{ MHz}$$

(c) Vehicle is moving perpendicular to the angle of arrival of the transmitted signal

$\theta = 90^\circ$ $\cos \theta = \cos 90^\circ = 0$ there is no Doppler shift received frequency = transmitted frequency.

Example 1.3 Compute the RMS delay spread for power delay profile



(b) If BPSK modulation is used, what is the maximum bit rate that can be sent through the channel?

Solution:

(a) RMS delay spread $\sigma_\tau = \sqrt{\tau^{-2} - (\bar{\tau})^2}$

$$\begin{aligned} \text{Mean excess delay } \bar{\tau} &= \frac{\sum_K P(\tau_K) \tau_K}{\sum_K P(\tau_K)} \\ &= \frac{1(0) + 1(1)}{1 + 1} = \frac{1}{2} \\ &= 0.5 \mu s \end{aligned}$$

$$\begin{aligned} \tau^{-2} &= \frac{\sum_K P(\tau_K) \tau_K^2}{\sum_K P(\tau_K)} \\ &= \frac{1(0)^2 + 1(1)^2}{1 + 1} \\ &= \frac{1}{2} = 0.5 \mu sec \\ \sigma_\tau &= \sqrt{\tau^{-2} - (\bar{\tau})^2} \\ &= \sqrt{0.5 - (0.5)^2} = 0.5 \mu sec \end{aligned}$$

(b) Bit rate R_b = symbol rate R_S

$$R_S = \frac{1}{T_S}$$

$$\frac{\sigma_\tau}{T_S} \leq 0.1$$

$$T_S \geq \frac{\sigma_\tau}{0.1} = \frac{0.5 \mu s}{0.1} = 5 \mu s$$

$$R_S = \frac{1}{T_S}$$

$$= \frac{1}{5 \times 10^6} = 200 \text{ kbps}$$

$$\text{Bit rate } R_b = 200 \text{ kbps}$$

$$\begin{aligned} \text{(c) Coherence bandwidth } B_C &= \frac{1}{5 \sigma_\tau} \\ &= \frac{1}{5(0.5 \times 10^{-6})} = 4 \text{ MHz} \end{aligned}$$

Example 1.4 Calculate coherence time, Doppler spread for carrier frequency $f_c = 1900 \text{ MHz}$ and $V = 50 \text{ m/s}$ of moving vehicle distance of 10 m .

Solution:

$$\begin{aligned} \text{(a) Coherence time } T_C &= \frac{9}{16 \pi f_m} = \frac{9 \lambda}{16 \pi V} = \frac{9 c}{16 \pi V f_c} \\ \text{Wavelength } \lambda &= \frac{c}{f_c} \\ V &= f_m \lambda \\ \text{Velocity } V &= f_m \lambda \\ f_m &= \frac{V}{\lambda} \\ T_C &= \frac{9 \times 3 \times 10^8}{16 \pi (50 \times 1900 \times 10^6)} = 565 \mu\text{s} \end{aligned}$$

$$\begin{aligned} \text{(b) Doppler spread } B_D = f_m &= \frac{V}{\lambda} = \frac{V f_c}{c} = \frac{50 \times 1900 \times 10^6}{3 \times 10^8} \\ B_D &= 316.66 \text{ Hz} \end{aligned}$$

Example 1.5 Find the Fraunhofer (far-field) distance for an antenna with maximum dimension of 1 m and operating frequency of 9000 MHz . If antennas have unity gain, calculate the path loss. [Nov 2013, April 2017, Nov 2012, Nov 2009]

Solution:

Operating frequency, $f = 900 \text{ MHz}$

$$\lambda = c/f = \frac{3 \times 10^8 \text{ m/s}}{900 \times 10^6} = 0.33 \text{ m}$$

$$\text{Fraunhofer distance, } d_f = \frac{2d^2}{\lambda} = \frac{2(1)^2}{0.33} = 6 \text{ m}$$

$$\text{Path loss } PL(\text{dB}) = -10 \log \left[\frac{\lambda}{(4\pi)^2 d^2} \right] = -10 \log \left[\frac{(0.33^2)}{(4 \times 3.14)^2 \cdot 6^2} \right] = 47 \text{ dB}$$

Example 1.6 If a transmitter produces 50 W of power, express the transmit power in units of (a) dBm , and (b) dBW . If 50 W is applied to a unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 1000 m from the antenna. What is $P_r(10 \text{ km})$? Assume unity gain for the receiver antenna.

Solution:

Given:

$$\text{Transmitter power, } P_t = 50 \text{ W}$$

Carrier frequency, $f_c = 900 \text{ MHz}$

$$\text{Using equation } P_r(d) \text{ dBm} = 10 \log \left[\frac{P_t(d_o)}{0.001 \text{ W}} \right] + 20 \log \left(\frac{d_o}{d} \right)$$

$$(a) \text{ Transmitter power, } P_t(d) \text{ dBm} = 10 \log \left[\frac{P_t(mW)}{1 \text{ mW}} \right] = 10 \log [50 \times 10^3] = 47.0 \text{ dBm}$$

$$(b) \text{ Transmitter power, } P_t(dBW) = 10 \log \left[\frac{P_t(W)}{1 \text{ W}} \right] = 10 \log [50] = 17.0 \text{ dBW}$$

The received power can be determined using equation

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L} = \frac{50(1)(1)(1/3)^2}{(4\pi)^2 (100^2)(1)} = (5 \times 10^{-6}) \text{ W} = 3.5 \times 10^{-3} \text{ mW}$$

$$P_r \text{ (dBm)} = 10 \log P_r \text{ (mW)} = 10 \log (3.5 \times 10^{-3} \text{ mW}) = -24.5 \text{ dBm}$$

The received power at 10m can be expressed in terms of dBm using equation

$$P_r(d) \text{ dBm} = 10 \log \left[\frac{P_r(d_o)}{0.001 \text{ W}} \right], \text{ where } d_o = 100 \text{ m and } d = 10 \text{ km.}$$

$$P_r(10 \text{ km}) = P_r(100) + 20 \log \left[\frac{100}{10000} \right]$$

$$= -24.5 \text{ dBm} - 40 \text{ dB}$$

$$= -64.5 \text{ dBm}$$

Example 1.6 Determine the proper spatial sampling interval required to make small scale propagation measurement which assume that consecutive samples are highly correlated in time. How many samples will be required over 10m travel distance if $f_c = 1900 \text{ MHz}$ and $v = 50 \text{ m/s}$. How long would it take to make these measurements, assuming they could be made in real time from a moving vehicle? What is the Doppler spread B_D for the channel? [April 2017]

Solution:

For correlation, ensure that the time between samples is equal to $T_c / 2$, and use the smallest value of T_c for conservative design.

$$\text{Using equation } T_c \approx \frac{9}{16\pi f_m}$$

$$T_c \approx \frac{9}{16\pi f_m} = \frac{9\lambda}{16\pi v} = \frac{9c}{16\pi v f_c}$$

$$= \frac{9 \times 3 \times 10^8}{16 \times 3.14 \times 50 \times 1900 \times 10^6} = 565 \text{ } \mu\text{s}$$

Taking time samples at less than half, at $282.5 T_c$ corresponds to a spatial sampling interval of

$$\Delta x = \frac{v T_c}{2} = \frac{50 \times 565 \mu\text{s}}{2} = 0.014125 \text{ m} = 1.41 \text{ cm}$$

Therefore, the number of samples required over a 10m travel distance is

$$N_x = \frac{10}{\Delta x} = \frac{100}{0.014125} = 701 \text{ samples}$$

The time taken to make this measurement is equal to $\frac{10}{50 \text{ ms}} = 0.2 \text{ s}$

$$\text{The Doppler spread is } B_D = f_m = \frac{v f_c}{c} = \frac{50 \times 1900 \times 10^6}{3 \times 10^8} = 316.66 \text{ Hz}$$

1.5. Consider a transmitter which radiates a sinusoidal carrier frequency of 1850 MHz. For a vehicle moving 60 mph, compute the received carrier frequency if the mobile is moving directly toward the transmitter. [April/May 2018] → (24)

Given Data:

Carrier
frequency
 $f_c =$
1850MHz
Vehicle
speed $V =$
60 mph

Solution:

Doppler shift is positive for vehicle is moving
towards the transmitter Received
frequency $f = f_c + f_d$

$$= 1850 \times 10^6 + \frac{V}{\lambda}$$

$$= 1850 \times 10^6 + \frac{100 \times 60 \times 60}{0.15}$$

$$= 2002 \text{ MHz}$$

$$f = \frac{c}{\lambda}$$

$$\lambda = \frac{c}{f}$$

$$= \frac{3 \times 10^8}{2002 \times 10^6}$$

$$= 0.15 \text{ m}$$

LINK BUDGET DESIGN USING PATH LOSS MODELS

- ✓ Radio propagation models are derived using analytical and empirical (**observed**) methods.
- ✓ Empirical approach:
 - It is based on fitting curves or analytical expressions that recreate a set of measured data.
 - It takes all propagation factors in account, both known and unknown, through actual field measurements.
- ✓ Some classical propagation models have emerged to predict large scale coverage for mobile communication systems design.
- ✓ In mobile communication systems, **pathloss models** are used **to calculate received signal level** as a function of distance, to predict the SNR.
- ✓ Some **practical path loss estimation techniques** are
 - Log-distance Path Loss Model
 - Log-normal Shadowing
 - Determination of percentage coverage area.
 -

a. Log-distance path loss model

- ✓ Both analytical and measurement based models indicate that average received signal power decreases logarithmically with distance (whether in outdoor or indoor channels).
- ✓ The average large-scale path loss for an arbitrary T-R separation is expressed as a function of distance d using the pathloss component, 'n'

$$PL(d) \propto \left(\frac{d}{d_0} \right)^n \rightarrow (1)$$

in dB,

$$PL(dB) = \overline{PL}(d_0) + 10 n \log \left(\frac{d}{d_0} \right) \rightarrow (2)$$

where

$\overline{PL}(d)$	→	Average large-scale path loss
d	→	T-R separation distance
d_0	→	Close-in reference distance (determined from measurements close to the transmitter)
n	→	Path loss exponent (the rate at which the pathloss increases)

- ✓ Path loss exponent n , varies with environment.

Eg: $n=2$ for free space

Table 1.1: Path loss exponents for different environments

Environment	Path Loss Exponent, n
Free space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building line-of-sight	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factories	2 to 3

- ✓ Reference distance d_0 ,
 - For large cellular system → $d_0=1\text{Km}$
 - For Microcellular system → $d_0=100\text{m (or) } 1\text{m}$ [i.e., much smaller]

b. Log-normal Shadowing

- ✓ The log-distance path loss model in equation (2) does not consider the fact that the environmental may be vastly different at two different locations having the same T-R separation.
- ✓ The path loss $PL(d)$ at a particular location is random and distributed log-normally (in dB) about the mean distance-dependent value.

$$PL(d) \text{ [dB]} = \overline{PL}(d) + X_\sigma \rightarrow (3)$$

$$= PL(d_0) + 10n \log \left(\frac{d}{d_0} \right) + X_\sigma \rightarrow (4)$$

$$\text{Received power} = \text{Transmitted power} - \text{Path loss} \rightarrow (5)$$

$$P_r(d) \text{ [dBm]} = P_t \text{ [dBm]} - PL(d) \text{ [dB]} \rightarrow (6)$$

where $PL(d) \rightarrow$ Path loss at a particular location

$P_r(d) \rightarrow$ Normal distribution about $PL(d) \rightarrow$ Received power

$X_\sigma \rightarrow$ Zero-mean Gaussian distributed random variable (in dB)

$\sigma \rightarrow$ Standard deviation (in dB)

- ✓ d_0, n, σ determine the path loss.
- ✓ The log-normal distribution describes the random *shadowing* effects occur over a large number of locations with the same T-R separation, but with different levels of clutter on the propagation path. This phenomenon is referred to as log-normal shadowing.

Q function (or) error function (erf)

- ✓ Q function is used to determine the probability that received signal strength will exceed a particular level.

$$Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^\infty \exp\left(-\frac{x^2}{2}\right) dx = \frac{1}{2} \left[1 - \text{erf}\left(\frac{z}{\sqrt{2}}\right) \right] \rightarrow (7)$$

$$\text{where } Q(z) = 1 - Q(-z) \rightarrow (8)$$

- ✓ Probability that received signal level will exceed a certain value γ can be calculates as,

$$\Pr[P_r(d) > \gamma] = Q\left(\frac{\gamma - P_r(d)}{\sigma}\right) \rightarrow (9)$$

- ✓ Similarly, the probability that received signal level will be below γ is given by,

$$\Pr[P_r(d) < \gamma] = Q\left(\frac{P_r(d) - \gamma}{\sigma}\right) \rightarrow (10)$$

where, $\gamma \rightarrow$ certain value of received power

c. Determination of percentage coverage area

- ✓ Due to random effects of shadowing, some locations within a coverage area will be below a particular desired received signal threshold.
- ✓ It is useful to compute the percentage of coverage area in boundary coverage.
- ✓ For a circular coverage area of radius R from the base station and the desired signal threshold of γ , the percentage of useful threshold area $U(\gamma)$ can be found by

$$U(\gamma) = \frac{1}{\pi R^2} \int \Pr[P_r(r) > \gamma] dA = \frac{1}{\pi R^2} \int_0^{2\pi} \int_0^R \Pr[P_r(r) > \gamma] dr d\theta$$

where,

$\Pr[P_r(r) > \gamma] \rightarrow$ Probability that the random received signal at $d = r$ exceeds the threshold – within an incremental area dA .

$d = r \rightarrow$ It represents the radial distance from the transmitter.

- ✓ For large number of values of σ and n
- $$U(\gamma) = \frac{1}{2} \left[1 + \exp\left(-\frac{1}{b^2}\right) \left(1 - \operatorname{erf}\left(\frac{1}{b}\right)\right) \right]$$

where,

$$b = (10 n \log e) / \sigma$$

$$\sqrt{2}$$

- ✓ $U(\gamma)$ is the percentage useful service area where received signal strength equals or greater than γ i.e. $P_R(d) > \gamma$.
- ✓ If $n=2$ and $\sigma = 8$ dB, a 75% boundary coverage provides 91% area coverage.
- ✓ If $n=3$ and $\sigma = 9$ dB, a 50% boundary coverage provides 71% area coverage

In wireless communication, **LCR** (Level-Crossing Rate) and **ADF** (Average Duration of Fades) are statistical quantities used to characterize the fading channel's behavior. The LCR measures how often a signal's envelope drops below a specified threshold, while the ADF indicates the average time the signal remains below that threshold. Together, they provide insights into the channel's quality, which is crucial for designing robust wireless systems, evaluating error burst statistics, and selecting appropriate channel coding schemes and interleaver sizes.

Level-Crossing Rate (LCR)

- **Definition:** The LCR quantifies the average number of times a radio signal's envelope crosses a given threshold level, typically in the negative-going direction.
- **Significance:** It offers insight into the "burstiness" or fluctuations of the fading channel. A higher LCR suggests more rapid changes in signal strength.

Average Duration of Fades (ADF)

- **Definition:** The ADF measures the average time interval during which the received signal's envelope remains below a specified threshold level.
- **Significance:** It is critical for analyzing error burst statistics, as longer fade durations can lead to longer periods of data corruption, affecting the system's overall reliability.

Why LCR and ADF are Important

- **Channel Characterization:** LCR and ADF are considered second-order statistics that provide more detailed information than simpler first-order statistics like the Probability Density Function (PDF) and Cumulative Distribution Function (CDF).
- **System Performance Analysis:** They are essential for evaluating the performance of wireless communication systems, particularly in the context of diversity techniques (like Selection Combining or Maximal Ratio Combining).
- **Error Control Design:** The knowledge of LCR and ADF helps in:
- **Channel Coding:** Choosing effective channel coding schemes to combat fading.
- **Interleaving:** Determining the optimal interleaver size to spread out error bursts.
- **Outage Rate:** Computing the outage rate and average outage duration, which directly relates to the reliability of the communication link.
- **Estimating Mobile Velocity:** The LCR can also be used to estimate the velocity of a mobile unit within the fading channel.